

Documentation v3

Revised Academic-Year / Batch Tracking Model

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System Architecture

Academic Structure

School



■■■ Department (Optional)



■■■ Programme



■■■ Batch

■■■ Semester

■ ■■■ Course

■ ■■■ Course Outcome (CO)

■■■ Students

Academic Year is the primary user-facing selector. Batch is maintained in the background for batch-wise tracking, student records, ATR continuity and longitudinal reports.

Programme defines duration and total semesters. Semester year is derived from semester number: 1–2 = Year 1, 3–4 = Year 2, 5–6 = Year 3, 7–8 = Year 4.

Module 1: Academic Management

Module Structure

Academic Management

■■■ School

■■■ Department

■■■ Programme

■■■ Academic Year

■■■ Batch

■■■ Semester

■■■ Course

■■■ Student

Module: Academic Management

Entity: School

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
code	VARCHAR(20)	No	Yes	-	School Code
name	VARCHAR(150)	No	Yes	-	School Name
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Soft Delete Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
School → Department	Department	One-to-Many	Yes
School → Programme	Programme	One-to-Many	Yes

3. API Contract

API	Method	Endpoint	Description
Create School	POST	/api/v1/schools	Create new school
Update School	PUT	/api/v1/schools/{id}	Update school
Get School	GET	/api/v1/schools/{id}	Get school
Get All Schools	GET	/api/v1/schools	Get all schools
Get Active Schools	GET	/api/v1/schools/active	Get active schools
Delete School	DELETE	/api/v1/schools/{id}	Delete/Deactivate school

4. Validation

- code is required and unique.
- name is required and unique.
- status must be ACTIVE or INACTIVE.

5. Business Rules

Rule No.	Rule
BR-01	School code must be unique.
BR-02	School name must be unique.
BR-03	A School may contain multiple Departments and Programmes.
BR-04	A School cannot be hard deleted when referenced.

Module: Academic Management

Entity: Department

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
schoolId	BIGINT	Yes	FK	NULL	References School
code	VARCHAR(20)	No	Yes*	-	Department Code
name	VARCHAR(100)	No	Yes*	-	Department Name
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Soft Delete Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Department → School	School	Many-to-One	Yes
Department → Programme	Programme	One-to-Many	Yes

3. API Contract

API	Method	Endpoint	Description
Create Department	POST	/api/v1/departments	Create department
Update Department	PUT	/api/v1/departments/{id}	Update department
Get Department	GET	/api/v1/departments/{id}	Get department
Get All Departments	GET	/api/v1/departments	Get all departments
Get by School	GET	/api/v1/schools/{schoolId}/departments	Get departments by school
Delete Department	DELETE	/api/v1/departments/{id}	Delete/Deactivate department

4. Validation

- code is required and unique in its School when applicable.
- name is required and unique in its School when applicable.
- schoolId is optional.
- status must be ACTIVE or INACTIVE.

5. Business Rules

Rule No.	Rule
BR-01	Department is optional.
BR-02	A Programme may or may not belong to a Department.
BR-03	Department cannot be hard deleted if referenced by a Programme.

Module: Academic Management

Entity: Programme

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
departmentId	BIGINT	Yes	FK	NULL	References Department
code	VARCHAR(20)	No	Yes	-	Programme Code
name	VARCHAR(150)	No	Yes	-	Programme Name
shortName	VARCHAR(20)	Yes	No	NULL	Abbreviation
durationYears	INTEGER	No	No	-	Programme Duration
totalSemesters	INTEGER	No	No	-	Total Semesters
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Soft Delete Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Programme → Department	Department	Many-to-One	Yes
Programme → Batch	Batch	One-to-Many	No
Programme → PO	PO	One-to-Many	No
Programme → PSO	PSO	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create Programme	POST	/api/v1/programmes	Create programme
Update Programme	PUT	/api/v1/programmes/{id}	Update programme
Get Programme	GET	/api/v1/programmes/{id}	Get programme
Get All Programmes	GET	/api/v1/programmes	Get programmes
Get by Department	GET	/api/v1/departments/{departmentId}/programmes	Get department programmes
Delete Programme	DELETE	/api/v1/programmes/{id}	Delete/Deactivate programme

4. Validation

- code and name are required and unique.
- durationYears > 0.
- totalSemesters > 0.
- departmentId is optional.

5. Business Rules

Rule No.	Rule
BR-01	Programme defines duration and total number of semesters.
BR-02	Semester year is derived: 1–2 Year 1, 3–4 Year 2, 5–6 Year 3, 7–8 Year 4.
BR-03	Programme cannot be hard deleted when referenced.

Module: Academic Management

Entity: Academic Year

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
name	VARCHAR(20)	No	Yes	-	YYYY-YY
startDate	DATE	No	No	-	Start Date
endDate	DATE	No	No	-	End Date
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Academic Year → Batch	Batch	One-to-Many	No
Academic Year → Semester	Semester	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create Academic Year	POST	/api/v1/academic-years	Create academic year
Update Academic Year	PUT	/api/v1/academic-years/{id}	Update academic year
Get Academic Year	GET	/api/v1/academic-years/{id}	Get academic year
Get All Academic Years	GET	/api/v1/academic-years	Get academic years
Get Active Academic Years	GET	/api/v1/academic-years/active	Get active academic years
Delete Academic Year	DELETE	/api/v1/academic-years/{id}	Delete/Deactivate academic year

4. Validation

- name must follow YYYY-YY.
- startDate required.
- endDate > startDate.

5. Business Rules

Rule No.	Rule
BR-01	Academic Year name must be unique.
BR-02	Only one Academic Year can be ACTIVE at a time.
BR-03	Academic Year periods must not overlap.
BR-04	Academic Year is the primary user-facing selector.
BR-05	Batch is resolved in the background from Programme + Academic Year + semester progression.

Module: Academic Management

Entity: Batch

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
programmeId	BIGINT	No	FK	-	References Programme
code	VARCHAR(30)	No	Yes	-	Batch Identifier
startYear	INTEGER	No	No	-	Start Year
endYear	INTEGER	No	No	-	End Year
status	ENUM(ACTIVE, COMPLETED, ARCHIVED)	No	No	ACTIVE	Lifecycle Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Batch → Programme	Programme	Many-to-One	No
Batch → Academic Year	Academic Year	Many-to-One	No
Batch → Semester	Semester	One-to-Many	No
Batch → Student	Student	One-to-Many	No
Batch → Course	Course	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create Batch	POST	/api/v1/batches	Create batch
Update Batch	PUT	/api/v1/batches/{id}	Update batch
Get Batch	GET	/api/v1/batches/{id}	Get batch
Get Programme Batches	GET	/api/v1/programmes/{programmeId}/batches	Get programme batches
Update Batch Status	PUT	/api/v1/batches/{id}/status	Change batch lifecycle

4. Validation

- programmeId must exist.
- code unique within Programme.
- endYear > startYear.
- status must be ACTIVE, COMPLETED or ARCHIVED.

5. Business Rules

Rule No.	Rule
BR-01	HOD controls batch lifecycle.
BR-02	Completed batches remain available for history and reports.
BR-03	Archived batches are read-only for normal academic operations.
BR-04	Batch is maintained even when users work through Academic Year.

Module: Academic Management

Entity: Semester

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
batchId	BIGINT	No	FK	-	References Batch
academicYearId	BIGINT	No	FK	-	References Academic Year
semesterNumber	INTEGER	No	No	-	Semester Number
startDate	DATE	No	No	-	Start Date
endDate	DATE	No	No	-	End Date
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Semester → Batch	Batch	Many-to-One	No
Semester → Academic Year	Academic Year	Many-to-One	No
Semester → Course	Course	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create Semester	POST	/api/v1/semesters	Create semester
Update Semester	PUT	/api/v1/semesters/{id}	Update semester
Get Semester	GET	/api/v1/semesters/{id}	Get semester
Get Batch Semesters	GET	/api/v1/batches/{batchId}/semesters	Get batch semesters
Delete Semester	DELETE	/api/v1/semesters/{id}	Delete/Deactivate semester

4. Validation

- batchId and academicYearId must exist.
- semesterNumber must be positive and within Programme totalSemesters.
- endDate > startDate.

5. Business Rules

Rule No.	Rule
BR-01	Semester belongs to Batch and Academic Year.
BR-02	Semester number is unique within a Batch.
BR-03	Semester year is derived from semester number.
BR-04	Semester cannot be hard deleted when Course or assessment data exist.

Module: Academic Management

Entity: Course

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
batchId	BIGINT	No	FK	-	References Batch
semesterId	BIGINT	No	FK	-	References Semester
code	VARCHAR(20)	No	Yes*	-	Course Code
name	VARCHAR(200)	No	No	-	Course Name
credits	DECIMAL(3,1)	No	No	-	Credits
type	ENUM(THEORY, PRACTICAL, THEORY_PRACTICAL, PROJECT, AUDIT)	No	No	-	Course Type
courseCoordinatorId	BIGINT	Yes	FK	NULL	References User
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Course → Batch	Batch	Many-to-One	No
Course → Semester	Semester	Many-to-One	No
Course → CO	Course Outcome	One-to-Many	No
Course → User	User	Many-to-One	Yes

3. API Contract

API	Method	Endpoint	Description
Create Course	POST	/api/v1/courses	Create course
Update Course	PUT	/api/v1/courses/{id}	Update course
Get Course	GET	/api/v1/courses/{id}	Get course
Get Batch Courses	GET	/api/v1/batches/{batchId}/courses	Get batch courses
Get Semester Courses	GET	/api/v1/semesters/{semesterId}/courses	Get semester courses
Delete Course	DELETE	/api/v1/courses/{id}	Delete/Deactivate course

4. Validation

- batchId and semesterId must exist.
- name and code required.
- credits > 0.
- semester must belong to selected Batch.

5. Business Rules

Rule No.	Rule
BR-01	HOD creates and manages Courses with name, code, semester and credits.
BR-02	Course code is unique in the applicable Batch/Programme context.
BR-03	Course cannot be hard deleted when CO, assessment or attainment data exist.

Module: Academic Management

Entity: Student

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
batchId	BIGINT	No	FK	-	References Batch
prn	VARCHAR(30)	No	Yes	-	Permanent Registration Number
name	VARCHAR(150)	No	No	-	Student Name
status	ENUM(ACTIVE, INACTIVE, PASSED, ARCHIVED)	No	No	ACTIVE	Student Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Student → Batch	Batch	Many-to-One	No
Student → Uploaded Student Marks	Uploaded Student Marks	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create Student	POST	/api/v1/students	Create student
Update Student	PUT	/api/v1/students/{id}	Update student
Get Student	GET	/api/v1/students/{id}	Get student
Get Students by Batch	GET	/api/v1/batches/{batchId}/students	Get batch students
Get Student by PRN	GET	/api/v1/students/prn/{prn}	Get student by PRN

4. Validation

- prn required and unique.
- name required.
- batchId must exist.

5. Business Rules

Rule No.	Rule
BR-01	Student records are resolved from direct-assessment uploads when PRN is not already present.
BR-02	Existing PRN records are reused.
BR-03	Student remains linked to Batch for longitudinal reports and ATR continuity.
BR-04	Historical student assessment data is not deleted.

Module 2: User & Access Management

Module Structure

- User & Access Management
 - ■ ■ User
 - ■ ■ Role / Assignment

Module: User & Access Management

Entity: User

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
username	VARCHAR(100)	No	Yes	-	Login Identifier
email	VARCHAR(150)	No	Yes	-	Email
passwordHash	VARCHAR(255)	No	No	-	Password Hash
role	ENUM(IQAC, DIRECTOR, HOD, PROGRAMME_COORDINATOR, COURSE_COORDINATOR, FACULTY)	No	No	-	Portal Role
programmeId	BIGINT	Yes	FK	NULL	Optional Programme Scope
departmentId	BIGINT	Yes	FK	NULL	Optional Department Scope
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	User Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
User → Programme	Programme	Many-to-One	Yes
User → Department	Department	Many-to-One	Yes

3. API Contract

API	Method	Endpoint	Description
Login	POST	/api/v1/auth/login	Login using username or email
Logout	POST	/api/v1/auth/logout	Logout
Create User	POST	/api/v1/users	Create user
Update User	PUT	/api/v1/users/{id}	Update user
Get User	GET	/api/v1/users/{id}	Get user
Get Users	GET	/api/v1/users	Get users

4. Validation

- Username/email and password required.
- Role must be supported.
- Referenced Programme/Department must exist when supplied.

5. Business Rules

Rule No.	Rule
BR-01	Login supports username or email.
BR-02	JWT authentication is used.
BR-03	Authorization is role based.
BR-04	Department scope remains optional.

Module: User & Access Management

Entity: Role / Assignment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
userId	BIGINT	No	FK	-	References User
scopeType	ENUM(DEPARTMENT, PROGRAMME, BATCH, COURSE)	No	No	-	Assignment Scope
scopeId	BIGINT	No	No	-	Scope ID
assignmentType	ENUM(HOD, PROGRAMME_COORDINATOR, COURSE_COORDINATOR)	No	No	-	Responsibility
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Assignment Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Assignment → User	User	Many-to-One	No
Assignment → Scope	Department / Programme / Batch / Course	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Create Assignment	POST	/api/v1/user-assignments	Assign responsibility
Update Assignment	PUT	/api/v1/user-assignments/{id}	Update assignment
Get User Assignments	GET	/api/v1/users/{userId}/assignments	Get assignments
Deactivate Assignment	DELETE	/api/v1/user-assignments/{id}	Deactivate assignment

4. Validation

- userId and scope must exist.
- assignmentType must be compatible with scope.

5. Business Rules

Rule No.	Rule
BR-01	HOD responsibility is Department scoped.
BR-02	Programme Coordinator responsibility is Programme scoped.
BR-03	Course Coordinator responsibility is Course scoped.
BR-04	Historical assignments are retained.

Module 3: Outcome Management

Module Structure

Outcome Management

■■■ Programme Outcome (PO)

■■■ PO Competency

■■■ Programme Specific Outcome (PSO)

■■■ PSO Competency

■■■ Course Outcome (CO)

Module: Outcome Management

Entity: Programme Outcome (PO)

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
programmeId	BIGINT	No	FK	-	References Programme
code	VARCHAR(20)	No	No	-	PO Code
statement	TEXT	No	No	-	PO Statement
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
PO → Programme	Programme	Many-to-One	No
PO → PO Competency	PO Competency	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create PO	POST	/api/v1/pos	Create PO
Update PO	PUT	/api/v1/pos/{id}	Update PO
Get PO	GET	/api/v1/pos/{id}	Get PO
Get Programme POs	GET	/api/v1/programmes/{programmeId}/pos	Get POs
Delete PO	DELETE	/api/v1/pos/{id}	Delete/Deactivate PO

4. Validation

- programmeId, code and statement required.
- Code unique within Programme.

5. Business Rules

Rule No.	Rule
BR-01	Every PO belongs to a Programme.
BR-02	PO count is dynamic.
BR-03	PO cannot be hard deleted when referenced.

Module: Outcome Management

Entity: PO Competency

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
poId	BIGINT	No	FK	-	References PO
statement	TEXT	No	No	-	Competency Statement
displayOrder	INTEGER	No	No	-	Display Sequence
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
PO Competency → PO	PO	Many-to-One	No
PO Competency → CO-PO Mapping	CO-PO Mapping	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create PO Competency	POST	/api/v1/po-competencies	Create competency
Update PO Competency	PUT	/api/v1/po-competencies/{id}	Update competency
Get by PO	GET	/api/v1/pos/{poId}/competencies	Get PO competencies
Delete	DELETE	/api/v1/po-competencies/{id}	Delete/deactivate

4. Validation

- poId and statement required.
- displayOrder must be positive.

5. Business Rules

Rule No.	Rule
BR-01	Competencies are dynamic.
BR-02	displayOrder is unique within a PO.
BR-03	Competency cannot be hard deleted when mapped.

Module: Outcome Management

Entity: Programme Specific Outcome (PSO)

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
programmeId	BIGINT	No	FK	-	References Programme
code	VARCHAR(20)	No	No	-	PSO Code
statement	TEXT	No	No	-	PSO Statement
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
PSO → Programme	Programme	Many-to-One	No
PSO → PSO Competency	PSO Competency	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create PSO	POST	/api/v1/psos	Create PSO
Update PSO	PUT	/api/v1/psos/{id}	Update PSO
Get PSO	GET	/api/v1/psos/{id}	Get PSO
Get Programme PSOs	GET	/api/v1/programmes/{programmeId}/psos	Get PSOs
Delete PSO	DELETE	/api/v1/psos/{id}	Delete/Deactivate PSO

4. Validation

- programmeId, code and statement required.
- Code unique within Programme.

5. Business Rules

Rule No.	Rule
BR-01	Every PSO belongs to a Programme.
BR-02	PSO count is dynamic.
BR-03	PSO cannot be hard deleted when referenced.

Module: Outcome Management

Entity: PSO Competency

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
psoid	BIGINT	No	FK	-	References PSO
statement	TEXT	No	No	-	Competency Statement
displayOrder	INTEGER	No	No	-	Display Sequence
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
PSO Competency → PSO	PSO	Many-to-One	No
PSO Competency → CO-PSO Mapping	CO-PSO Mapping	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create PSO Competency	POST	/api/v1/psos-competencies	Create competency
Update PSO Competency	PUT	/api/v1/psos-competencies/{id}	Update competency
Get by PSO	GET	/api/v1/psos/{psoid}/competencies	Get PSO competencies
Delete	DELETE	/api/v1/psos-competencies/{id}	Delete/deactivate

4. Validation

- psoid and statement required.
- displayOrder must be positive.

5. Business Rules

Rule No.	Rule
BR-01	Competencies are dynamic.
BR-02	displayOrder is unique within a PSO.
BR-03	Competency cannot be hard deleted when mapped.

Module: Outcome Management

Entity: Course Outcome (CO)

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseId	BIGINT	No	FK	-	References Course
code	VARCHAR(20)	No	No	-	CO Code
statement	TEXT	No	No	-	CO Statement
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
CO → Course	Course	Many-to-One	No
CO → CO-PO Mapping	CO-PO Mapping	One-to-Many	No
CO → CO-PSO Mapping	CO-PSO Mapping	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create CO	POST	/api/v1/courses/{courseId}/cos	Create CO
Update CO	PUT	/api/v1/cos/{id}	Update CO
Get CO	GET	/api/v1/cos/{id}	Get CO
Get Course COs	GET	/api/v1/courses/{courseId}/cos	Get COs
Delete CO	DELETE	/api/v1/cos/{id}	Delete/Deactivate CO

4. Validation

- courseId, code and statement required.
- CO code unique within Course.

5. Business Rules

Rule No.	Rule
BR-01	CO count is dynamic.
BR-02	CO cannot be hard deleted when mapping, assessment or attainment data exist.

Module 4: CO Mapping Management

Module Structure

CO Mapping Management

■■■ CO-P0 Competency Mapping

■■■ CO-PS0 Competency Mapping

Module: CO Mapping Management

Entity: CO-PO Competency Mapping

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseOutcomeId	BIGINT	No	FK	-	References CO
programmeOutcomeCompetencyId	BIGINT	No	FK	-	References PO Competency
mappingLevel	DECIMAL(3,2)	No	No	0	Mapping Strength
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Mapping → CO	CO	Many-to-One	No
Mapping → PO Competency	PO Competency	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Create Mapping	POST	/api/v1/mappings/co-po	Create CO-PO mapping
Update Mapping	PUT	/api/v1/mappings/co-po/{id}	Update mapping
Get Course Mappings	GET	/api/v1/mappings/co-po/course/{courseId}	Get course mappings
Delete Mapping	DELETE	/api/v1/mappings/co-po/{id}	Delete mapping

4. Validation

- CO and PO Competency must exist.
- Duplicate mapping is not allowed.
- Mapping level must be within allowed range.

5. Business Rules

Rule No.	Rule
BR-01	A CO may map to multiple PO competencies.
BR-02	Mappings may be created manually or imported from Excel.

Module: CO Mapping Management

Entity: CO-PSO Competency Mapping

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseOutcomeId	BIGINT	No	FK	-	References CO
programmeSpecificOutcomeCompetencyId	BIGINT	No	FK	-	References PSO Competency
mappingLevel	DECIMAL(3,2)	No	No	0	Mapping Strength
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Mapping → CO	CO	Many-to-One	No
Mapping → PSO Competency	PSO Competency	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Create Mapping	POST	/api/v1/mappings/co-psyo	Create CO-PSO mapping
Update Mapping	PUT	/api/v1/mappings/co-psyo/{id}	Update mapping
Get Course Mappings	GET	/api/v1/mappings/co-psyo/course/{courseId}	Get course mappings
Delete Mapping	DELETE	/api/v1/mappings/co-psyo/{id}	Delete mapping

4. Validation

- CO and PSO Competency must exist.
- Duplicate mapping is not allowed.
- Mapping level must be within allowed range.

5. Business Rules

Rule No.	Rule
BR-01	A CO may map to multiple PSO competencies.
BR-02	Mappings may be created manually or imported from Excel.

Module 5: Assessment Data Management

Module Structure

Assessment Data Management

■■■ End Semester Marks Upload

■■■ Uploaded Student Marks

■■■ Student CO Marks

■■■ Course End Survey

■■■ Survey Response

■■■ Survey Response Details

■■■ Programme Exit Survey

Module: Assessment Data Management

Entity: End Semester Marks Upload

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
uploadedBy	BIGINT	No	FK	-	User
originalFileName	VARCHAR(255)	No	No	-	Excel File
uploadStatus	ENUM(SUCCESS, FAILED, PROCESSING)	No	No	PROCESSING	Status
totalStudents	INTEGER	No	No	0	Students Processed
fileSize	BIGINT	No	No	-	File Size
processingTime	INTEGER	Yes	No	NULL	Processing Time
errorMessage	VARCHAR(255)	Yes	No	NULL	Error Message
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit
updatedAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Upload → Course	Course	Many-to-One	No
Upload → Batch	Batch	Many-to-One	No
Upload → Semester	Semester	Many-to-One	No
Upload → Academic Year	Academic Year	Many-to-One	No
Upload → User	User	Many-to-One	No
Upload → Uploaded Student Marks	Uploaded Student Marks	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Upload Marks	POST	/api/v1/marks/end-semester/upload	Upload End Semester Marks Excel
Get Uploads	GET	/api/v1/marks/end-semester/uploads	Get uploads
Get Upload	GET	/api/v1/marks/end-semester/uploads/{id}	Get upload details
Delete Upload	DELETE	/api/v1/marks/end-semester/uploads/{id}	Delete upload

4. Validation

- Course, Batch, Semester and Academic Year must exist.
- Excel must be valid and match active template.
- Total students > 0.

5. Business Rules

Rule No.	Rule
BR-01	Students are resolved by PRN and linked to Batch.
BR-02	Upload status changes PROCESSING → SUCCESS/FAILED.
BR-03	Upload metadata is retained for audit and recalculation.

Module: Assessment Data Management

Entity: Uploaded Student Marks

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
marksUploadId	BIGINT	No	FK	-	Upload
studentId	BIGINT	No	FK	-	Student
prn	VARCHAR(30)	No	No	-	PRN Snapshot
studentName	VARCHAR(150)	No	No	-	Name Snapshot
totalMarks	DECIMAL(6,2)	Yes	No	NULL	Total Marks
maxMarks	DECIMAL(6,2)	Yes	No	NULL	Maximum Marks
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Uploaded Student Marks → Upload	Marks Upload	Many-to-One	No
Uploaded Student Marks → Student	Student	Many-to-One	No
Uploaded Student Marks → Student CO Marks	Student CO Marks	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Get Student Marks	GET	/api/v1/marks/end-semester/uploads/{uploadId}/students/{studentId}	Get student marks
Get CO Marks	GET	/api/v1/marks/end-semester/uploads/{uploadId}/co-marks	Get CO marks

4. Validation

- Upload and Student must exist.
- PRN required.

5. Business Rules

Rule No.	Rule
BR-01	Created internally during upload processing.
BR-02	Existing Student is reused by PRN.

Module: Assessment Data Management

Entity: Student CO Marks

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
uploadedStudentMarksId	BIGINT	No	FK	-	Uploaded Student Marks
courseOutcomeId	BIGINT	No	FK	-	CO
obtainedMarks	DECIMAL(6,2)	No	No	0	Obtained Marks
maximumMarks	DECIMAL(6,2)	No	No	0	Maximum Marks
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Student CO Marks → Uploaded Student Marks	Uploaded Student Marks	Many-to-One	No
Student CO Marks → CO	CO	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get Student CO Marks	GET	/api/v1/marks/end-semester/students/{studentId}/co-marks	Get student CO marks
Get Course CO Marks	GET	/api/v1/marks/end-semester/courses/{courseId}/co-marks	Get course CO marks

4. Validation

- CO must belong to selected Course.
- Obtained $\geq$ 0.
- Obtained $\leq$ Maximum.
- Maximum $>$ 0.

5. Business Rules

Rule No.	Rule
BR-01	Each record represents one student for one CO.
BR-02	Source data for Direct CO Attainment.

Module: Assessment Data Management

Entity: Course End Survey

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
uploadedBy	BIGINT	No	FK	-	User
originalFileName	VARCHAR(255)	No	No	-	Excel File
uploadStatus	ENUM(SUCCESS, FAILED, PROCESSING)	No	No	PROCESSING	Status
totalResponses	INTEGER	No	No	0	Responses
errorMessage	VARCHAR(255)	Yes	No	NULL	Error Message
fileSize	BIGINT	No	No	-	File Size
processingTime	INTEGER	Yes	No	NULL	Processing Time
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Survey → Course	Course	Many-to-One	No
Survey → Batch	Batch	Many-to-One	No
Survey → Semester	Semester	Many-to-One	No
Survey → Academic Year	Academic Year	Many-to-One	No
Survey → Survey Response	Survey Response	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Upload Survey	POST	/api/v1/surveys/course-end/upload	Upload Course End Survey
Get Surveys	GET	/api/v1/surveys/course-end	Get uploads
Get Survey	GET	/api/v1/surveys/course-end/{id}	Get survey
Delete Survey	DELETE	/api/v1/surveys/course-end/{id}	Delete survey

4. Validation

- Course, Batch, Semester and Academic Year must exist.
- Excel must match active template.
- Responses > 0.

5. Business Rules

Rule No.	Rule
BR-01	Used for Indirect CO Attainment.
BR-02	Upload status changes PROCESSING → SUCCESS/FAILED.
BR-03	Historical upload metadata is retained.

Module: Assessment Data Management

Entity: Survey Response

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
surveyId	BIGINT	No	FK	-	Course End Survey
studentId	BIGINT	Yes	FK	NULL	Optional Student
responseCode	VARCHAR(50)	No	No	-	Response Identifier
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Response → Survey	Course End Survey	Many-to-One	No
Response → Student	Student	Many-to-One	Yes
Response → Details	Survey Response Details	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Get Responses	GET	/api/v1/surveys/course-end/{surveyId}/responses	Get responses

4. Validation

· surveyId and responseCode required.

5. Business Rules

Rule No.	Rule
BR-01	Responses are created from survey upload.
BR-02	Anonymous responses are supported when the template has no identity field.

Module: Assessment Data Management

Entity: Survey Response Details

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
surveyResponseId	BIGINT	No	FK	-	Survey Response
courseOutcomeId	BIGINT	Yes	FK	NULL	CO
responseCategory	ENUM(SLIGHT, MODERATE, SUBSTANTIAL)	No	No	-	Category
responseWeight	DECIMAL(5,2)	No	No	-	Configured Weight
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Detail → Response	Survey Response	Many-to-One	No
Detail → CO	CO	Many-to-One	Yes

3. API Contract

API	Method	Endpoint	Description
Get Response Details	GET	/api/v1/surveys/responses/{responseId}/details	Get details

4. Validation

- response must exist.
- Category must be supported.
- Weight resolved from active configuration.

5. Business Rules

Rule No.	Rule
BR-01	Weights are configuration-driven.
BR-02	Details are used for Indirect CO Attainment.

Module: Assessment Data Management

Entity: Programme Exit Survey

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
uploadedBy	BIGINT	No	FK	-	User
originalFileName	VARCHAR(255)	No	No	-	Excel File
uploadStatus	ENUM(SUCCESS, FAILED, PROCESSING)	No	No	PROCESSING	Status
totalResponses	INTEGER	No	No	0	Responses
errorMessage	VARCHAR(255)	Yes	No	NULL	Error Message
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Exit Survey → Programme	Programme	Many-to-One	No
Exit Survey → Batch	Batch	Many-to-One	No
Exit Survey → Academic Year	Academic Year	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Upload Exit Survey	POST	/api/v1/surveys/programme-exit/upload	Upload Programme Exit Survey
Get Exit Surveys	GET	/api/v1/surveys/programme-exit	Get surveys
Get Exit Survey	GET	/api/v1/surveys/programme-exit/{id}	Get survey
Delete Exit Survey	DELETE	/api/v1/surveys/programme-exit/{id}	Delete survey

4. Validation

- Programme, Batch and Academic Year must exist.
- Excel must be valid.
- Responses > 0.

5. Business Rules

Rule No.	Rule
BR-01	Separate from Course End Survey.
BR-02	Associated with the completing Batch.
BR-03	Used for indirect Programme PO/PSO attainment.

Module 6: Configuration Management

Module Structure

Configuration Management

■■■ Attainment Configuration

■■■ Attainment Level Configuration

■■■ Excel Template Configuration

Module: Configuration Management

Entity: Attainment Configuration

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
directAttainmentWeight	DECIMAL(5,2)	No	No	80.00	Direct Weight %
indirectAttainmentWeight	DECIMAL(5,2)	No	No	20.00	Indirect Weight %
directAttainmentThreshold	DECIMAL(5,2)	No	No	60.00	Student Threshold %
slightResponseWeight	DECIMAL(5,2)	No	No	0.33	Slight Weight
moderateResponseWeight	DECIMAL(5,2)	No	No	0.66	Moderate Weight
substantialResponseWeight	DECIMAL(5,2)	No	No	1.00	Substantial Weight
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Configuration → Calculation Engines	CO / PO / PSO Engines	Used By	No

3. API Contract

API	Method	Endpoint	Description
Create	POST	/api/v1/configurations/attainment	Create configuration
Update	PUT	/api/v1/configurations/attainment/{id}	Update configuration
Get Active	GET	/api/v1/configurations/attainment	Get active configuration
Get	GET	/api/v1/configurations/attainment/{id}	Get configuration

4. Validation

- Direct + Indirect weight = 100.
- Threshold 0–100.
- Response weights >= 0.
- Only one ACTIVE configuration.

5. Business Rules

Rule No.	Rule
BR-01	Active configuration applies to future calculations.
BR-02	Historical configurations are retained.
BR-03	Only authorized administrators modify configuration.

Module: Configuration Management

Entity: Attainment Level Configuration

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
level	INTEGER	No	Yes	-	Attainment Level
minimumPercentage	DECIMAL(5,2)	No	No	-	Minimum %
maximumPercentage	DECIMAL(5,2)	No	No	-	Maximum %
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Level Configuration → CO Engine	CO Attainment Engine	Used By	No
Level Configuration → PO/PSO Engine	PO/PSO Engine	Used By	No

3. API Contract

API	Method	Endpoint	Description
Create Level	POST	/api/v1/configurations/attainment-levels	Create level
Update Level	PUT	/api/v1/configurations/attainment-levels/{id}	Update level
Get Levels	GET	/api/v1/configurations/attainment-levels	Get levels
Delete Level	DELETE	/api/v1/configurations/attainment-levels/{id}	Delete level

4. Validation

- Level > 0.
- 0 <= min <= max <= 100.
- Active ranges must not overlap.

5. Business Rules

Rule No.	Rule
BR-01	Example configuration: <50 = 1, 50–70 = 2, >70 = 3.
BR-02	Historical configurations are retained.

Module: Configuration Management

Entity: Excel Template Configuration

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
templateName	VARCHAR(100)	No	Yes	-	Template Name
templateType	ENUM(END_SEMESTER_MARKS, COURSE_END_SURVEY, PROGRAMME_EXIT_SURVEY)	No	No	-	Template Type
sheetName	VARCHAR(100)	No	No	-	Sheet
headerRowNumber	INTEGER	No	No	1	Header Row
configuration	JSON	No	No	-	Column Mapping
status	ENUM(ACTIVE, INACTIVE)	No	No	ACTIVE	Status
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Template → Upload Process	Marks / Survey Uploads	Used By	No

3. API Contract

API	Method	Endpoint	Description
Create Template	POST	/api/v1/configurations/excel-templates	Create template
Update Template	PUT	/api/v1/configurations/excel-templates/{id}	Update template
Get Templates	GET	/api/v1/configurations/excel-templates	Get templates
Get Active	GET	/api/v1/configurations/excel-templates/type/{templateType}	Get active template
Delete Template	DELETE	/api/v1/configurations/excel-templates/{id}	Delete/deactivate

4. Validation

- Template name unique.
- Supported type required.
- Sheet name nonblank.
- Header row > 0.
- Configuration JSON valid.
- One ACTIVE template per type.

5. Business Rules

Rule No.	Rule
BR-01	Changes affect future uploads only.
BR-02	Historical templates retained for audit and rollback.

Module 7: CO Attainment Engine

Module Structure

CO Attainment Engine
■■■ Calculation Run
■■■ Direct CO Attainment
■■■ Indirect CO Attainment
■■■ Overall CO Attainment

Module: CO Attainment Engine

Entity: Calculation Run

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
triggeredBy	BIGINT	No	FK	-	User
status	ENUM(PENDING, RUNNING, SUCCESS, FAILED)	No	No	PENDING	Status
startedAt	TIMESTAMP	No	No	Current Timestamp	Start
completedAt	TIMESTAMP	Yes	No	NULL	Completion
remarks	TEXT	Yes	No	NULL	Failure / Notes

2. Relationships

Relationship	Entity	Type	Nullable
Run → Course	Course	Many-to-One	No
Run → Batch	Batch	Many-to-One	No
Run → Semester	Semester	Many-to-One	No
Run → Academic Year	Academic Year	Many-to-One	No
Run → User	User	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Start Calculation	POST	/api/v1/attainments/calculate	Start calculation
Get Runs	GET	/api/v1/attainments/calculation-runs	Get calculation runs
Get Run	GET	/api/v1/attainments/calculation-runs/{id}	Get details

4. Validation

- Course, Batch, Semester, Academic Year and User must exist.
- Only one run for same Course + Batch + Semester + Academic Year may execute at a time.

5. Business Rules

Rule No.	Rule
BR-01	Every execution creates a new run.
BR-02	Status PENDING → RUNNING → SUCCESS/FAILED.
BR-03	All generated records reference the run.
BR-04	Previous runs remain available for audit.

Module: CO Attainment Engine

Entity: Direct CO Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
configurationId	BIGINT	No	FK	-	Attainment Configuration
courseOutcomeId	BIGINT	No	FK	-	CO
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
attainedStudents	INTEGER	No	No	0	Attained Students
totalStudents	INTEGER	No	No	0	Total Students
attainmentPercentage	DECIMAL(5,2)	No	No	0	Percentage
attainmentLevel	INTEGER	No	No	-	Level
calculatedAt	TIMESTAMP	No	No	Current Timestamp	Calculation Time

2. Relationships

Relationship	Entity	Type	Nullable
Direct CO → Student CO Marks	Student CO Marks	Calculated From	No
Direct CO → Calculation Run	Calculation Run	Many-to-One	No
Direct CO → CO	CO	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get Direct	GET	/api/v1/attainments/direct	Get direct results
Get Course Direct	GET	/api/v1/attainments/direct/course/{courseId}	Get course results
Get Direct	GET	/api/v1/attainments/direct/{id}	Get result

4. Validation

- CO and configuration must exist.
- Total Students > 0.

5. Business Rules

Rule No.	Rule
BR-01	Student attained when marks >= configured threshold.
BR-02	Percentage is converted to configured level.
BR-03	Level, not raw percentage, is used in Overall CO.

Module: CO Attainment Engine

Entity: Indirect CO Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
configurationId	BIGINT	No	FK	-	Attainment Configuration
courseOutcomeId	BIGINT	No	FK	-	CO
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
weightedScore	DECIMAL(8,2)	No	No	0	Weighted Score
attainmentPercentage	DECIMAL(5,2)	No	No	0	Percentage
attainmentLevel	INTEGER	No	No	-	Level

2. Relationships

Relationship	Entity	Type	Nullable
Indirect CO → Course End Survey	Course End Survey	Calculated From	No
Indirect CO → Calculation Run	Calculation Run	Many-to-One	No
Indirect CO → CO	CO	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get Indirect	GET	/api/v1/attainments/indirect	Get indirect results
Get Course Indirect	GET	/api/v1/attainments/indirect/course/{courseId}	Get course results
Get Indirect	GET	/api/v1/attainments/indirect/{id}	Get result

4. Validation

- Survey and CO must exist.
- Response weights must be configured.

5. Business Rules

Rule No.	Rule
BR-01	Weighted Score uses Slight/Moderate/Substantial weights.
BR-02	Percentage is converted to level.
BR-03	Level is used in Overall CO.

Module: CO Attainment Engine

Entity: Overall CO Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
configurationId	BIGINT	No	FK	-	Attainment Configuration
courseOutcomeId	BIGINT	No	FK	-	CO
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
directAttainmentLevel	DECIMAL(5,2)	No	No	0	Direct Level
indirectAttainmentLevel	DECIMAL(5,2)	No	No	0	Indirect Level
overallAttainment	DECIMAL(5,2)	No	No	0	Final Overall CO

2. Relationships

Relationship	Entity	Type	Nullable
Overall CO → Direct CO	Direct CO Attainment	Many-to-One	No
Overall CO → Indirect CO	Indirect CO Attainment	Many-to-One	No
Overall CO → Calculation Run	Calculation Run	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get Overall CO	GET	/api/v1/attainments/overall-co	Get overall CO results
Get Course Overall CO	GET	/api/v1/attainments/overall-co/course/{courseId}	Get course results
Get Overall CO	GET	/api/v1/attainments/overall-co/{id}	Get result

4. Validation

- Direct and Indirect records must exist.
- Configured weights must total 100.

5. Business Rules

Rule No.	Rule
BR-01	Overall CO = Direct Level × Direct Weight + Indirect Level × Indirect Weight.
BR-02	Calculation uses attainment levels, not percentages.

Module 8: PO & PSO Attainment Engine

Module Structure

PO & PSO Attainment Engine

■■■ PO Competency Attainment

■■■ PO Attainment

■■■ PSO Competency Attainment

■■■ PSO Attainment

■■■ Programme Aggregation

Module: PO & PSO Attainment Engine

Entity: PO Competency Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
programmeOutcomeCompetencyId	BIGINT	No	FK	-	PO Competency
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
attainmentValue	DECIMAL(5,2)	No	No	0	Average Overall CO
mappedCourseOutcomes	INTEGER	No	No	0	Mapped CO Count
calculatedAt	TIMESTAMP	No	No	Current Timestamp	Calculation Time

2. Relationships

Relationship	Entity	Type	Nullable
PO Competency Attainment → PO Competency	PO Competency	Many-to-One	No
PO Competency Attainment → Overall CO	Overall CO	Calculated From	No

3. API Contract

API	Method	Endpoint	Description
Get PO Competencies	GET	/api/v1/attainments/po-competencies	Get results
Get PO Competency	GET	/api/v1/attainments/po-competencies/{id}	Get result
Get Programme	GET	/api/v1/attainments/po-competencies/programme/{programmeId}	Get programme results

4. Validation

- PO Competency must exist.
- At least one mapped CO required.

5. Business Rules

Rule No.	Rule
BR-01	Average of mapped Overall CO Attainments.
BR-02	Uses same Calculation Run.
BR-03	Historical calculations remain unchanged.

Module: PO & PSO Attainment Engine

Entity: PO Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
programmeOutcomeId	BIGINT	No	FK	-	PO
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
attainmentValue	DECIMAL(5,2)	No	No	0	PO Attainment
totalCompetencies	INTEGER	No	No	0	Competencies
calculatedAt	TIMESTAMP	No	No	Current Timestamp	Calculation Time

2. Relationships

Relationship	Entity	Type	Nullable
PO Attainment → PO	PO	Many-to-One	No
PO Attainment → PO Competency Attainment	PO Competency Attainment	Calculated From	No
PO Attainment → Batch	Batch	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get PO Results	GET	/api/v1/attainments/pos	Get results
Get PO	GET	/api/v1/attainments/pos/{id}	Get result
Get Batch PO	GET	/api/v1/attainments/pos/batch/{batchId}	Get batch results

4. Validation

- PO must exist.
- Competency results must exist.

5. Business Rules

Rule No.	Rule
BR-01	Calculated from PO Competency Attainment.
BR-02	Batch and Academic Year are retained.
BR-03	Historical results are preserved.

Module: PO & PSO Attainment Engine

Entity: PSO Competency Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
programmeSpecificOutcomeCompetencyId	BIGINT	No	FK	-	PSO Competency
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
attainmentValue	DECIMAL(5,2)	No	No	0	Average Overall CO
mappedCourseOutcomes	INTEGER	No	No	0	Mapped CO Count
calculatedAt	TIMESTAMP	No	No	Current Timestamp	Calculation Time

2. Relationships

Relationship	Entity	Type	Nullable
PSO Competency Attainment → PSO Competency	PSO Competency	Many-to-One	No
PSO Competency Attainment → Overall CO	Overall CO	Calculated From	No

3. API Contract

API	Method	Endpoint	Description
Get PSO Competencies	GET	/api/v1/attainments/pso-competencies	Get results
Get PSO Competency	GET	/api/v1/attainments/pso-competencies/{id}	Get result
Get Programme	GET	/api/v1/attainments/pso-competencies/programme/{programmeId}	Get programme results

4. Validation

- PSO Competency must exist.
- At least one mapped CO required.

5. Business Rules

Rule No.	Rule
BR-01	Average of mapped Overall CO Attainments.
BR-02	Uses same Calculation Run.

Module: PO & PSO Attainment Engine

Entity: PSO Attainment

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	No	FK	-	Calculation Run
programmeSpecificOutcomeId	BIGINT	No	FK	-	PSO
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
attainmentValue	DECIMAL(5,2)	No	No	0	PSO Attainment
totalCompetencies	INTEGER	No	No	0	Competencies
calculatedAt	TIMESTAMP	No	No	Current Timestamp	Calculation Time

2. Relationships

Relationship	Entity	Type	Nullable
PSO Attainment → PSO	PSO	Many-to-One	No
PSO Attainment → PSO Competency Attainment	PSO Competency Attainment	Calculated From	No
PSO Attainment → Batch	Batch	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get PSO Results	GET	/api/v1/attainments/psos	Get results
Get PSO	GET	/api/v1/attainments/psos/{id}	Get result
Get Batch PSO	GET	/api/v1/attainments/psos/batch/{batchId}	Get batch results

4. Validation

- PSO must exist.
- Competency results must exist.

5. Business Rules

Rule No.	Rule
BR-01	Calculated from PSO Competency Attainment.
BR-02	Batch and Academic Year are retained.
BR-03	Historical results are preserved.

Module: PO & PSO Attainment Engine

Entity: Programme Aggregation

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
directPOAttainment	DECIMAL(5,2)	No	No	0	Programme Direct PO
indirectPOAttainment	DECIMAL(5,2)	No	No	0	Programme Indirect PO
finalPOAttainment	DECIMAL(5,2)	No	No	0	Final PO
directPSOAttainment	DECIMAL(5,2)	No	No	0	Programme Direct PSO
indirectPSOAttainment	DECIMAL(5,2)	No	No	0	Programme Indirect PSO
finalPSOAttainment	DECIMAL(5,2)	No	No	0	Final PSO
calculatedAt	TIMESTAMP	No	No	Current Timestamp	Calculation Time

2. Relationships

Relationship	Entity	Type	Nullable
Aggregation → Programme	Programme	Many-to-One	No
Aggregation → Batch	Batch	Many-to-One	No
Aggregation → PO/PSO	PO / PSO Attainment	Aggregates	No
Aggregation → Exit Survey	Programme Exit Survey	Uses	No

3. API Contract

API	Method	Endpoint	Description
Calculate Programme Attainment	POST	/api/v1/programme-attainment/calculate	Calculate programme attainment
Get Programme Attainment	GET	/api/v1/programme-attainment/{id}	Get programme result
Get Batch Result	GET	/api/v1/programme-attainment/batch/{batchId}	Get batch result

4. Validation

- Programme and Batch must exist.
- Required course-level PO/PSO results must exist.
- Programme Exit Survey must exist for indirect results.

5. Business Rules

Rule No.	Rule
BR-01	Programme direct PO/PSO attainment is aggregated from course-level results.
BR-02	Programme Exit Survey provides indirect evidence.
BR-03	Final programme result uses configured direct/indirect weights.
BR-04	Batch history is retained for longitudinal reporting and ATR continuity.

Module 9: ATR & Approval Management

Module Structure

ATR & Approval Management

■■■ Course ATR

■■■ Programme ATR

■■■ Approval Request

■■■ Approval History

Module: ATR & Approval Management

Entity: Course ATR

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
courseId	BIGINT	No	FK	-	Course
batchId	BIGINT	No	FK	-	Batch
semesterId	BIGINT	No	FK	-	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
attainmentValue	DECIMAL(5,2)	No	No	0	Attainment
targetValue	DECIMAL(5,2)	No	No	0	Target
status	ENUM(DRAFT, SUBMITTED, REVISION_REQUIRED, APPROVED)	No	No	DRAFT	ATR Status
observation	TEXT	Yes	No	NULL	Observation
actionPlan	TEXT	Yes	No	NULL	Action Plan
createdBy	BIGINT	No	FK	-	User
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Course ATR → Course	Course	Many-to-One	No
Course ATR → Batch	Batch	Many-to-One	No
Course ATR → Semester	Semester	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Create Course ATR	POST	/api/v1/atr/course	Create Course ATR
Get Course ATR	GET	/api/v1/atr/course/{id}	Get ATR
Update Course ATR	PUT	/api/v1/atr/course/{id}	Update ATR
Submit Course ATR	PUT	/api/v1/atr/course/{id}/submit	Submit ATR

4. Validation

- Course attainment and target must exist.
- Approved ATR cannot be edited directly.

5. Business Rules

Rule No.	Rule
BR-01	ATR compares attainment with target.
BR-02	Observation and action plan are retained.
BR-03	Approval creates an audit trail.

Module: ATR & Approval Management

Entity: Programme ATR

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	No	FK	-	Batch
academicYearId	BIGINT	No	FK	-	Academic Year
finalPOAttainment	DECIMAL(5,2)	No	No	0	Final PO
finalPSOAttainment	DECIMAL(5,2)	No	No	0	Final PSO
targetPO	DECIMAL(5,2)	No	No	0	PO Target
targetPSO	DECIMAL(5,2)	No	No	0	PSO Target
status	ENUM(DRAFT, SUBMITTED, REVISION_REQUIRED, APPROVED)	No	No	DRAFT	ATR Status
observation	TEXT	Yes	No	NULL	Observation
actionPlan	TEXT	Yes	No	NULL	Action Plan
createdBy	BIGINT	No	FK	-	User
createdAt	TIMESTAMP	No	No	Current Timestamp	Audit

2. Relationships

Relationship	Entity	Type	Nullable
Programme ATR → Programme	Programme	Many-to-One	No
Programme ATR → Batch	Batch	Many-to-One	No
Programme ATR → Programme Aggregation	Programme Aggregation	Uses	No
Programme ATR → Approval Request	Approval Request	One-to-Many	No

3. API Contract

API	Method	Endpoint	Description
Create Programme ATR	POST	/api/v1/atr/programme	Create Programme ATR
Get Programme ATR	GET	/api/v1/atr/programme/{id}	Get ATR
Submit Programme ATR	PUT	/api/v1/atr/programme/{id}/submit	Submit ATR
Batch ATR History	GET	/api/v1/atr/programme/batch/{batchId}	Get ATR history

4. Validation

- Programme, Batch and Academic Year must exist.
- Programme attainment must exist.
- Targets must be available.

5. Business Rules

Rule No.	Rule
BR-01	Programme ATR is batch scoped.
BR-02	Uses final Programme PO/PSO attainment.
BR-03	Action-plan continuity is retained across Academic Years.
BR-04	HOD reviews submitted Programme ATR.
BR-05	Approved ATR is an immutable historical snapshot.

Module: ATR & Approval Management

Entity: Approval Request

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
entityType	ENUM(COURSE_ATR, PROGRAMME_ATR)	No	No	-	Entity Type
entityId	BIGINT	No	No	-	Entity ID
submittedBy	BIGINT	No	FK	-	Submitter
reviewedBy	BIGINT	Yes	FK	NULL	Reviewer
status	ENUM(PENDING, APPROVED, REVISION_REQUIRED, REJECTED)	No	No	PENDING	Approval Status
comments	TEXT	Yes	No	NULL	Comments
submittedAt	TIMESTAMP	No	No	Current Timestamp	Submission Time
reviewedAt	TIMESTAMP	Yes	No	NULL	Review Time

2. Relationships

Relationship	Entity	Type	Nullable
Approval Request → User	User	Many-to-One	No
Approval Request → ATR	Course ATR / Programme ATR	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get Pending Approvals	GET	/api/v1/approvals/pending	Get pending approvals
Review Approval	PUT	/api/v1/approvals/{id}/review	Approve or request revision
Get Approval	GET	/api/v1/approvals/{id}	Get approval

4. Validation

- Submitted entity must exist.
- Reviewer must have authority.

5. Business Rules

Rule No.	Rule
BR-01	HOD reviews Programme ATR.
BR-02	Revision returns ATR to responsible coordinator.
BR-03	Status transitions are recorded in history.

Module: ATR & Approval Management

Entity: Approval History

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
approvalRequestId	BIGINT	No	FK	-	Approval Request
fromStatus	VARCHAR(50)	No	No	-	Previous Status
toStatus	VARCHAR(50)	No	No	-	New Status
actionBy	BIGINT	No	FK	-	User
comments	TEXT	Yes	No	NULL	Comments
actionAt	TIMESTAMP	No	No	Current Timestamp	Action Time

2. Relationships

Relationship	Entity	Type	Nullable
History → Approval Request	Approval Request	Many-to-One	No
History → User	User	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Get Approval History	GET	/api/v1/approvals/{id}/history	Get approval history

4. Validation

· approvalRequestId and actionBy must exist.

5. Business Rules

Rule No.	Rule
BR-01	Every status transition is recorded.
BR-02	History is append-only.
BR-03	History supports audit and ATR traceability.

Module 10: Reports

Module Structure

- Reports
- Report Generation

Module: Reports

Entity: Report Generation

1. Entity Definition

Field	Data Type	Nullable	Unique	Default	Remarks
id	BIGINT	No	PK	Auto Generated	Primary Key
calculationRunId	BIGINT	Yes	FK	NULL	Calculation Run
generatedBy	BIGINT	No	FK	-	User
programmeId	BIGINT	No	FK	-	Programme
batchId	BIGINT	Yes	FK	NULL	Batch
semesterId	BIGINT	Yes	FK	NULL	Semester
academicYearId	BIGINT	No	FK	-	Academic Year
reportType	ENUM(CO_ATTAINMENT, PO_ATTAINMENT, PSO_ATTAINMENT, PROGR AMME_ATTAINMENT, COURSE_ATR, PROGRAMME_ATR, CONSOLIDATED)	No	No	-	Report Type
version	INTEGER	No	No	1	Report Version
fileName	VARCHAR(255)	No	No	-	Generated File
filePath	VARCHAR(500)	No	No	-	Storage Path
fileFormat	ENUM(PDF, EXCEL)	No	No	PDF	Format
generatedParameters JSON	JSON	No	No	-	Generation Filters
generatedAt	TIMESTAMP	No	No	Current Timestamp	Generation Time

2. Relationships

Relationship	Entity	Type	Nullable
Report → Calculation Run	Calculation Run	Many-to-One	Yes
Report → User	User	Many-to-One	No
Report → Programme	Programme	Many-to-One	No
Report → Batch	Batch	Many-to-One	Yes
Report → Semester	Semester	Many-to-One	Yes
Report → Academic Year	Academic Year	Many-to-One	No

3. API Contract

API	Method	Endpoint	Description
Generate Report	POST	/api/v1/reports/generate	Generate report
Get Reports	GET	/api/v1/reports	Get generated reports
Get Report	GET	/api/v1/reports/{id}	Get report details
Download Report	GET	/api/v1/reports/download/{id}	Download report
Delete Report	DELETE	/api/v1/reports/{id}	Delete report record

4. Validation

- Programme and Academic Year must exist.
- Batch required for batch-scoped reports.
- Report Type and File Format must be valid.

5. Business Rules

Rule No.	Rule
BR-01	Academic Year is the primary report filter.
BR-02	Batch is included for historical and longitudinal views.
BR-03	Multiple versions are allowed.

Rule No.	Rule
BR-04	Parameters are stored for reproducibility.
BR-05	Generated reports are snapshots and do not change after new calculations.

Attainment Calculation Methodology

Direct CO Attainment

- Direct CO Percentage = (Students attaining the configured threshold / Total students) × 100.
- The percentage is converted to a configured attainment level.
- The stored Direct CO Attainment used in the final calculation is the attainment level, not the raw percentage.

Indirect CO Attainment

- Weighted Score = (Slight × 0.33) + (Moderate × 0.66) + (Substantial × 1.00), using the active configuration.
- The weighted result is converted to a percentage and then to an attainment level.
- The stored Indirect CO Attainment used in the final calculation is the attainment level.

Overall CO Attainment

- Overall CO = (Direct Level × Direct Weight) + (Indirect Level × Indirect Weight).
- Default example: Direct 80%, Indirect 20%.
- Example: Direct Level 3 and Indirect Level 2 → 2.8.
- Overall Course CO Attainment = average of Overall CO values for all COs of the course.

PO / PSO Attainment

- PO Competency Attainment = average of mapped Overall CO Attainment values.
- PO Attainment = aggregation of PO competency attainments.
- PSO Competency Attainment follows the same mapped-CO logic.
- PSO Attainment = aggregation of PSO competency attainments.

Programme Attainment

- Programme Direct PO/PSO Attainment is aggregated from course-level results for the Programme and Batch.
- Programme Exit Survey provides indirect PO/PSO evidence.
- Final Programme Attainment = Direct × configured Direct Weight + Indirect × configured Indirect Weight.
- Final programme results are compared with configured targets for ATR.

Complete Business Workflows

Academic Structure Flow

Select Academic Year.
System resolves Programme, Batch and applicable Semester.
HOD manages active, completed and archived Batches.
HOD creates Courses with name, code, semester and credits.
Students are resolved from assessment uploads by PRN and linked to Batch.
Reports use Academic Year as the primary selector and Batch for history.

Course Attainment Flow

Upload End Semester Marks.
Upload Course End Survey.
Validate both files against active Excel templates.
Create Calculation Run.
Calculate Direct CO Attainment.
Calculate Indirect CO Attainment.
Calculate Overall CO Attainment.
Calculate PO/PSO Competency Attainment.
Calculate PO/PSO Attainment.
Store results against the Calculation Run.

Programme Attainment Flow

Collect course-level PO/PSO results.
Aggregate to Programme Direct PO/PSO Attainment.
Upload Programme Exit Survey.
Calculate Programme Indirect PO/PSO Attainment.
Calculate final Programme PO/PSO Attainment.
Compare against targets.
Generate Programme ATR.

ATR Approval Flow

Programme Coordinator prepares ATR.
ATR is submitted for review.
HOD reviews.
HOD approves or requests revision.
Revision returns to the responsible coordinator.
Approval History records every status transition.
Approved ATR becomes a historical snapshot.

Complete System Flow

Academic Year Selection
↓
Programme / Batch / Semester Resolution
↓
Course Management
↓
Student Resolution from Assessment Upload
↓
CO / PO / PSO Mapping
↓
End Semester Marks + Course End Survey
↓
Calculation Run
↓
Direct + Indirect CO Attainment
↓
Overall CO Attainment
↓
PO/PSO Competency + PO/PSO Attainment
↓
Programme Aggregation + Programme Exit Survey

↓
Final Programme PO/PSO Attainment
↓
Target Comparison
↓
Course / Programme ATR
↓
HOD Review / Approval
↓
PDF / Excel Reports

Key Design Rules

- Academic Year is the main selector shown to users.
- Batch is maintained in the background for historical tracking, student records, reports and ATR continuity.
- Department remains optional.
- Programme defines duration and total semesters.
- Semester year is derived from semester number.
- HOD manages active, completed and archived Batches.
- HOD creates Courses with name, code, semester and credits.
- Students are resolved by PRN during direct-assessment upload and linked to Batch.
- Course End Survey is used for Indirect CO Attainment.
- Programme Exit Survey is used for Indirect Programme PO/PSO Attainment.
- Attainment calculations are versioned by Calculation Run.
- Historical calculations and approved ATR snapshots are not modified by later calculations.
- Reports are snapshots and retain their generation parameters.

Documentation Status

This revised Documentation v3 follows the supplied PDF format: module structure, entity definition, relationships, API contract, validation and business rules. Existing v3 attainment methodology is retained where applicable; the finalized academic-year-driven batch tracking, student continuity, programme aggregation, ATR and approval workflow are incorporated.